A Counterexample to Question 4 on Dispersed Subspaces of L_p

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Status

This packet records a candidate counterexample, likely valid, to Question 4 of Gonzalez–Martinez-Abejon–Martinon [1]. The question asks whether, for $2 < p < \infty$, every dispersed subspace M of $L_p(0,1)$ satisfies

$$\beta(Q_M) = 1,$$

where $Q_M : L_p(0,1) \to L_p(0,1)/M$ is the quotient map and

$$\beta(T) = \inf \left\{ \liminf_{n\to\infty} \|Tx_n\| : \ (x_n) \subset S_E \text{ is disjoint} \right\}.$$

Source Question

The source paper proves the equality for $1 < p < 2$, observes that the analogous statement fails for $p = 2$, and then asks the following.

Proposition 3.12. *Let $1 < p < 2$ and let M be a dispersed subspace of L_p. Then the quotient map $Q_M : L_p \to L_p/M$ satisfies $\beta(Q_M) = 1$.*

Proof. We showed in Proposition 3.5 that, for $1 \le p < \infty$, a closed subspace L_p is strongly embedded if and only if it is dispersed. Moreover, for $1 < p < 2$, it was proved in [4, Theorem 2] that the unit ball B_M of a strongly embedded subspace of L_p is equi-integrable in L_p; i.e., that for each $\varepsilon > 0$ there exists $\delta > 0$ such that if $f \in B_M$ and A is a measurable subset of $(0,1)$ with $\mu(A) < \delta$ then $\|f \cdot \chi_A\|_p < \varepsilon$.

If $f \in L_p$ and $\|f\|_p = 1$ then $\|Q_M f\| = \mathrm{dist}(f, M) = \inf_{g \in 3B_M} \|f - g\|_p$. Thus, given a normalized disjoint sequence (f_n) in L_p and $g \in 3B_M$, and denoting $A_n = D(f_n)$, since $\lim_{n\to\infty} \mu(A_n) = 0$, there exists n_ε so that

$$\|f_n - g\|_p \ge \|f_n - g \cdot \chi_{A_n}\|_p \ge 1 - \varepsilon$$

for $n \ge n_\varepsilon$; hence $\liminf_{n\to\infty} \|Q_M f_n\| = 1$ and we get $\beta(Q_M) = 1$. $\square$

Remark 3.13. Proposition 3.12 fails for $p = 2$.

Indeed, if we consider the decomposition $L_2 = M \oplus M^\perp$ with both M and $M^\perp$ strongly embedded given in Proposition 3.4, then $\beta(Q_M) > 0$ and $\beta(Q_{M^\perp}) > 0$. Moreover, for $f \in L_2$ we have $\|f\|_2^2 = \mathrm{dist}(f, M)^2 + \mathrm{dist}(f, M^\perp)^2$, which implies $\beta(Q_M)^2 + \beta(Q_{M^\perp})^2 \le 1$. Hence $0 < \beta(Q_M), \beta(Q_{M^\perp}) < 1$.

Question 4. *Let M be a dispersed subspace of L_p $(2 < p < \infty)$. Is $\beta(Q_M) = 1$?*

For $2 < p < \infty$ there are strongly embedded subspaces of L_p whose unit ball is not equi-integrable in L_p [4]. So the argument in the proof of Proposition 3.12 is not valid.

Next we give further characterizations of DN-S(L_p, Y) and some consequences.

Theorem 3.14. *For $T \in L(L_p, Y)$, $1 < p < \infty$, the following assertions are equivalent:*

(1) *T is disjointly non-singular.*

Transcription of the relevant question from PDF page 9:

Question 4. Let M be a dispersed subspace of L_p $(2 < p < \infty)$. Is $\beta(Q_M) = 1$?

Proof Intuition

The $p < 2$ proof in the source uses equi-integrability of the unit ball of a strongly embedded subspace. For $p > 2$, one can hide a fixed part of every disjoint spike inside a Hilbertian subspace without making that subspace contain an almost-disjoint sequence. The trick is to attach each spike to a small Rademacher component living on a separate region. The Rademacher part forces the closed span to be isomorphic to ℓ_2, hence dispersed in L_p by the source paper's characterization, while the spike part keeps each normalized disjoint indicator at distance at most a prescribed number < 1 from the subspace.

Theorem 1. *Let $2 < p < \infty$ and let $0 < \eta < 1$. There is a dispersed closed subspace M of $L_p(0,1)$ such that*

$$\beta(Q_M) \le \eta.$$

In particular, Question 4 of [1] has a negative answer for every $2 < p < \infty$.

Proof. Split $(0,1)$ into two measurable pieces A and B of positive measure. Choose pairwise disjoint measurable sets $A_n \subset A$ with $\mu(A_n) > 0$, and put

$$f_n = \mu(A_n)^{-1/p}\mathbf{1}_{A_n}.$$

Then (f_n) is a normalized disjoint sequence in $L_p(0,1)$.

Let (r_n) be the usual Rademacher functions on B, normalized with respect to the probability measure on B, and set $u_n = \mu(B)^{-1/p} r_n \mathbf{1}_B$. Thus $\|u_n\|_p = 1$, and by the Khintchine inequality there are constants $0 < c_p \le C_p < \infty$ such that for every finitely supported scalar sequence (α_n),

$$c_p \left(\sum |\alpha_n|^2\right)^{1/2} \le \left\|\sum \alpha_n u_n\right\|_p \le C_p \left(\sum |\alpha_n|^2\right)^{1/2}.$$

Choose numbers $a, b > 0$ with $b/a \le \eta$, and define

$$g_n = af_n + bu_n, \qquad M = \overline{\operatorname{span}}\{g_n : n \in \mathbb{N}\}.$$

For every finitely supported (α_n), the supports of the spike part and the Rademacher part are disjoint, while the f_n's are mutually disjoint. Hence

$$\left\|\sum \alpha_n g_n\right\|_p^p = a^p \sum |\alpha_n|^p + b^p \left\|\sum \alpha_n u_n\right\|_p^p.$$

Since $p > 2$, $\|(\alpha_n)\|_{\ell_p} \le \|(\alpha_n)\|_{\ell_2}$. The preceding identity and Khintchine's inequality give

$$bc_p\|(\alpha_n)\|_{\ell_2} \le \left\|\sum \alpha_n g_n\right\|_p \le (a^p + b^p C_p^p)^{1/p}\|(\alpha_n)\|_{\ell_2}.$$

Thus (g_n) is equivalent to the unit vector basis of ℓ_2, and M is isomorphic to ℓ_2.

By Proposition 3.4 of [1], for $p > 2$ every closed infinite-dimensional subspace of L_p is strongly embedded if and only if it is isomorphic to ℓ_2. By Proposition 3.5 of the same paper, strongly embedded subspaces of L_p are exactly the dispersed subspaces. Therefore M is dispersed.

It remains to estimate $\beta(Q_M)$. For each n,

$$a^{-1}g_n = f_n + \frac{b}{a}u_n \in M.$$

Consequently

$$\operatorname{dist}(f_n, M) \le \left\|f_n - a^{-1}g_n\right\|_p = \frac{b}{a}\|u_n\|_p = \frac{b}{a} \le \eta.$$

Since (f_n) is a normalized disjoint sequence,

$$\beta(Q_M) \le \liminf_{n\to\infty} \|Q_M f_n\| = \liminf_{n\to\infty} \operatorname{dist}(f_n, M) \le \eta.$$

Taking any $\eta < 1$ gives a dispersed M with $\beta(Q_M) < 1$, contradicting the proposed equality. $\square$

Verification Notes

No numerical computation is used. The proof is purely analytic. The only external inputs are the Khintchine inequality and the source paper's Propositions 3.4 and 3.5, which identify ℓ_2-isomorphic subspaces of L_p, $p > 2$, as strongly embedded and identify strongly embedded subspaces with dispersed subspaces.

The construction gives the stronger quantitative statement $\inf\{\beta(Q_M) : M \subset L_p \text{ dispersed}\} = 0$ in the sense that for every $\eta > 0$ the packet constructs M with $\beta(Q_M) \le \eta$. This does not contradict the source paper's Theorem 3.14, which implies $\beta(Q_M) > 0$ for each fixed dispersed $M \subset L_p$.

Novelty Check

The cheap run indexes were searched for the arXiv id and the phrases

$$\texttt{disjointly non-singular}, \quad \texttt{DNS}, \quad \texttt{beta(Q\_M)}.$$

The only adjacent run hit before this packet was an older serious attempt on the related LSS/DSS problem for arXiv:1410.4752.

Local source search for $\beta(Q_M)$ found the source paper's positive $p < 2$ statement and Question 4, plus arXiv:2108.01052 citing only the positive $p < 2$ case. Later DNS papers arXiv:2101.06566 and arXiv:2302.04514 were inspected for the DNS/uniform-DNS and dispersed-subspace questions; they provide order-continuous positive results and later directions, but this bounded check did not find a $p > 2$ quotient counterexample.

Web searches for exact phrases around "Disjointly non-singular operators", "beta(Q_M)", and "strongly embedded" surfaced arXiv:2101.06566 and arXiv:2302.04514 as the relevant later DNS papers, with no explicit answer to Question 4 found in the search snippets. Novelty confidence is therefore bounded-search only, not a literature-complete guarantee.

Human Review Recommendation

Recommended for expert review as a likely valid counterexample. The reviewer should check especially the use of the source paper's characterization of strongly embedded subspaces for $p > 2$, and the quotient-distance estimate $\mathrm{dist}(f_n, M) \leq b/a$.

References

[1] M. Gonzalez, A. Martinez-Abejon, and A. Martinon, *Disjointly non-singular operators on Banach lattices*, arXiv:2012.10683, 2020.

[2] E. Bilokopytov, *Disjointly non-singular operators on order continuous Banach lattices complement the unbounded norm topology*, arXiv:2101.06566, 2021.

[3] M. Gonzalez and A. Martinon, *Disjointly non-singular operators: Extensions and local variations*, arXiv:2302.04514, 2023.